BRAC UNIVERSITY

Inspiring Excellence

CSE471: System Analysis and Design
Assignment on Functional Requirements
Proposed Project Title: Pet Care and Vet-Connect System

[Sample]

Group No: ##, CSE471 Lab Section: ##, [Semester] [Year]		
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Submission Date:

Project Overview

The "Pet Care and Vet-Connect System" is a comprehensive web platform designed to bridge the gap between pet owners and veterinary clinics. The system will manage three distinct user roles: Pet Owners, Vet Clinics, and Administrators. It will facilitate seamless appointment booking, provide a secure platform for managing pet health information, and offer premium features such as an AI-powered chatbot and a loyalty rewards program, all supported by a robust payment and notification system.

Tech Stack:

- Language: Typescript , Javascript
- Framework: Next.js
- Styling: TailwindCSS
- Database: PostgreSQL
- ORM: Prisma
- Deployment: Vercel
- External APIs:
    - **OpenAI API:** For AI chatbot functionality.
    - **Gmail API :** For sending notifications and reminders.
    - **Bkash API :** For processing subscription payments.

User Roles

1. **Pet Owner:** The primary users - they can register, create profiles for their pets, search for veterinarians, book appointments, write reviews, and access the AI chatbot (if premium).
2. **Vet Clinic:** A verified business - they can manage their clinic profile, add/manage doctor profiles, set appointment schedules, and manage incoming appointments (approve, cancel, reschedule).
3. **Admin:** A system super-user - responsible for verifying vet clinic registrations, monitoring system activity, and managing subscription/reward system parameters.

Functional Requirements

SL	Common Workflows
1	Registration and Login System: Users can create an account by providing basic details such as name, email, phone number, and password. Vet clinics complete a short multi-step form that includes their clinic name, location, and license number for verification. Once registered, both pet owners and clinics can log in securely to access their dashboards.
2	Admin Role & Access Management: An authorized administrator signs in to a dedicated panel to review pending vet clinic applications, check submitted details (e.g., clinic name, location, license), and approve or reject them. Only approved clinics appear in search and become bookable for pet owners.

Module 1	
Member	Feature Description
1	Vet Clinic Dashboard & Info Management: Verified clinics get a single dashboard to maintain their clinic profile and manage multiple doctor profiles. Clinics provide each doctor's name, email, base fee, qualifications, and rating, and doctors set their weekly availability and appointment slots (e.g., Mon 9–11 AM, 2–4 PM) so pet owners can book only open times.
2	Search & Appointment Booking System: Pet owners can find the right vet through a simple search and filter page—by location, clinic name, or rating—then open clinic and doctor profiles to see details, reviews, and available time slots. To book, owners choose which pet, pick a date and time, and confirm the appointment; any unavailable times are clearly shown as "Slot Booked."
3	Pet Information Management: Pet owners get a personal dashboard to keep their details up to date and manage their pets in one place. Owners can update name, location, and password, and create or edit multiple pet profiles with name, age, breed, gender, and vaccination status so vets have accurate information before appointments.

Module 2

Member	Feature Description
1	Vet Analytics Reports: Each clinic gets a private dashboard to track its performance over time. Vets can view total appointments by month or year, see estimated revenue based on completed appointments' base fees, and monitor their average rating trends. Clinics can filter by date ranges to spot busy periods and identify opportunities to improve service. For example, total appointments are counted based on completed visits, and each doctor's revenue is calculated from their completed appointments. The clinic's total revenue is then the combined total of all doctors' earnings. The average rating is found by dividing the total stars received by the number of reviews.
2	AI Chatbot (PetBot): Premium subscribers get 24/7 access to a smart assistant for general pet health and care questions. Ask about symptoms, diet, grooming, training, or routine care and receive instant, easy-to-follow guidance—plus suggestions on when to see a vet in person if something sounds urgent.
3	Review and Rating System: After a visit is marked Completed, pet owners can share feedback by giving the doctor a 1–5 star rating and writing a short review. These reviews help other owners choose confidently, and each doctor's profile shows an average rating calculated from all feedback.

Module 3	
Member	Feature Description
1	Vet Appointment Management: Clinics get a clear, date-sorted view of upcoming appointments and can approve, cancel, or reschedule with a few clicks. When an action is taken, the pet owner is notified, and rescheduling lets the clinic offer a new time slot so visits stay organized and on track.
1	Notifications & Email Reminders: Automatic emails keep everyone updated at each step of the appointment journey—booking confirmation to the pet owner, new booking alert to the clinic, status updates when an appointment is approved, canceled, or rescheduled (to the owner).
2	Subscription & Payment System: Users choose between Free and Premium plans. Free users can search, book appointments, and manage pets (the vet's service fee is paid directly to the clinic). Premium adds 24/7 PetBot access and

	loyalty discounts through reward points. The Premium subscription costs 200 Taka per month or 2,000 Taka per year, paid securely through bKash, with instant confirmation and an email receipt after purchase.
2	Reward Points System: Premium users earn loyalty points for key actions such as booking appointments and renewing subscriptions. These points convert into discounts on the next renewal, following a rate set by the admin (for example, 1,000 points = 50 ৳ discount). This feature rewards consistent users and encourages continued engagement with the platform.
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